

Module 5.1: Probability Models for Portfolio Return and Risk

SCHWESERNOTES - BOOK 1

LOS 5.a: Calculate and interpret the expected value, variance, standard deviation, covariances, and correlations of portfolio returns.

The **expected return of a portfolio** composed of n assets with weights, w_i , and expected returns, R_i , can be determined using the following formula:

$$E(R_P) = \sum_{i=1}^n w_i E(R_i) = w_1 E(R_1) + w_2 E(R_2) + \dots + w_n E(R_n)$$

The expected return and variance for a portfolio of assets can be determined using the properties of the individual assets in the portfolio. To do this, it is necessary to establish the portfolio weight for each asset. As indicated in the formula below, the weight, w_i , of Asset i is simply the market value currently invested in the asset divided by the current market value of the entire portfolio:

$$w_i = \frac{\text{market value of investment in Asset } i}{\text{market value of the portfolio}}$$

In many finance situations, we are interested in how two random variables move in relation to each other. For investment applications, one of the most frequently analyzed pairs of random variables is the returns of two assets. Investors and managers frequently ask questions such as, "What is the relationship between the return for Stock A and Stock B?" or, "What is the relationship between the performance of the S&P 500 and that of the automotive industry?"

Covariance is a measure of how two assets move together. It is the expected value of the product of the deviations of the two random variables from their respective expected values. A common symbol for the covariance between random variables X and Y is $\text{Cov}(X,Y)$. Because we will be mostly concerned with the covariance of asset returns, the following formula has been written in terms of the covariance of the return of Asset i , R_i , and the return of Asset j , R_j :

$$\text{Cov}(R_i, R_j) = E\{[R_i - E(R_i)][R_j - E(R_j)]\}$$

The following are *properties of covariance*:

- The covariance of a random variable with itself is its variance; that is, $\text{Cov}(R_A, R_A) = \text{Var}(R_A)$.
- Covariance may range from negative infinity to positive infinity.
- A positive covariance indicates that when one random variable is above its mean, the other random variable also tends to be above its mean.
- A negative covariance indicates that when one random variable is above its mean, the other random variable tends to be below its mean.

The **sample covariance** for a sample of returns data can be calculated as follows:

$$s_{X,Y} = \frac{\sum_{i=1}^n \{ [R_{1,i} - \bar{R}_1] [R_{2,i} - \bar{R}_2] \}}{n-1}$$

where:

$R_{1,i}$ = an observation of returns on Asset 1
 $R_{2,i}$ = an observation of returns on Asset 2
 $\bar{R}_1$ = mean return of Asset 1
 $\bar{R}_2$ = mean return of Asset 2
 n = number of observations in the sample

A **covariance matrix** shows the covariances between returns on a group of assets.

Covariance Matrix for Assets A, B, and C

Asset	A	B	C
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A	Cov(R_A, R_A)	Cov(R_A, R_B)	Cov(R_A, R_C)
B	Cov(R_B, R_A)	Cov(R_B, R_B)	Cov(R_B, R_C)
C	Cov(R_C, R_A)	Cov(R_C, R_B)	Cov(R_C, R_C)

Note that the diagonal terms from top left are the variances of each asset's returns—in other words, $\text{Cov}(R_A, R_A) = \text{Var}(R_A)$.

The covariance between the returns on two assets does not depend on order—in other words, $\text{Cov}(R_A, R_B) = \text{Cov}(R_B, R_A)$ —so in this covariance matrix, only three of the (off-diagonal) covariance terms are unique. In general for n assets, there are n variance terms (on the diagonal) and $n(n - 1) / 2$ unique covariance terms.

With **portfolio variance**, to calculate the variance of portfolio returns, we use the asset weights, returns variances, and returns covariances:

$$\text{Var}(R_P) = \sum_{i=1}^N \sum_{j=1}^N w_i w_j \text{Cov}(R_i, R_j)$$

The variance of a portfolio composed of two risky assets, A and B, can be expressed as follows:

$$\text{Var}(R_P) = w_A w_A \text{Cov}(R_A, R_A) + w_A w_B \text{Cov}(R_A, R_B) + w_B w_A \text{Cov}(R_B, R_A) + w_B w_B \text{Cov}(R_B, R_B)$$

We can write this more simply as:

$$\text{Var}(R_P) = w_A^2 \text{Var}(R_A) + w_B^2 \text{Var}(R_B) + 2w_A w_B \text{Cov}(R_A, R_B)$$

or:

$$\sigma_P^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \text{Cov}_{AB}$$

For a three-asset portfolio, the portfolio variance is:

$$\sigma_P^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + w_C^2 \sigma_C^2 + 2w_A w_B \text{Cov}_{AB} + 2w_A w_C \text{Cov}_{AC} + 2w_B w_C \text{Cov}_{BC}$$

Consider a portfolio with three assets: an index of domestic stocks (60%), an index of domestic bonds (30%), and an index of international equities (10%). A covariance matrix of the three assets is shown here.

Covariance Matrix for the Three Assets

Asset	Domestic Stocks	Domestic Bonds	International Equities
Domestic stocks	400	44	180
Domestic bonds	44	70	35
International equities	180	35	450

Portfolio returns variance =

$$(0.6^2)400 + (0.3^2)70 + (0.1^2)450 + 2(0.6)(0.3)44 + 2(0.6)(0.1)180 + 2(0.3)(0.1)35 = 194.34$$

Portfolio returns standard deviation = $\sqrt{194.34} = 13.94\%$

Note that the units of variance and covariance are %² (i.e., 0.001). When we put these values in as whole numbers (in %²), the portfolio variance is in %², and the standard deviation is in whole percentages. We could also put variance and covariance in as decimals and get both the portfolio returns variance and standard deviation as decimals.

From the formula for portfolio returns variance, we can see that the lower the covariance terms, the lower the portfolio variance (and standard deviation). This is true for positive values of covariance, as well as negative values.

Recall that the correlation coefficient for two variables is:

$$\rho_{AB} = \frac{\text{Cov}_{AB}}{\sigma_A \sigma_B}, \text{ so that } (\text{Cov}_{AB}) = \rho_{AB} \times \sigma_A \sigma_B$$

This can be substituted for Cov_{AB} in our formula for portfolio returns variance. With this substitution, we can use a correlation matrix to calculate portfolio returns variance, rather than using covariances.

Correlation Matrix for the Three Assets

Asset	Domestic Stocks	Domestic Bonds	International Equities
Domestic stocks	1.000	0.263	0.424
Domestic bonds	0.263	1.000	0.197
International equities	0.424	0.197	1.000

Note that the correlations of asset returns with themselves (the diagonal terms) are all 1.

LOS 5.b: Calculate and interpret the covariance and correlation of portfolio returns using a joint probability function for returns.

Example: Covariance of returns from a joint probability function

Assume that the economy can be in three possible states (S) next year: boom, normal, or slow economic growth. An expert source has calculated that $P(\text{boom}) = 0.30$, $P(\text{normal}) = 0.50$, and $P(\text{slow}) = 0.20$. The returns for Asset A, R_A , and Asset B, R_B , under each of the economic states are provided in the probability model as follows. What is the covariance of the returns for Asset A and Asset B?

Joint Probability Function

	$R_B = 30\%$	$R_B = 10\%$	$R_B = 0\%$
$R_A = 20\%$	0.30	0	0
$R_A = 12\%$	0	0.50	0
$R_A = 5\%$	0	0	0.20

The table gives us the joint probability of returns on Assets A and B (e.g., there is a 30% probability that the return on Asset A is 20% and the return on Asset B is 30%, and there is a 50% probability that the return on Asset A is 12% and the return on Asset B is 10%).

Answer:

First, we must calculate the expected returns for each of the assets:

$$E(R_A) = (0.3)(0.20) + (0.5)(0.12) + (0.2)(0.05) = 0.13$$

$$E(R_B) = (0.3)(0.30) + (0.5)(0.10) + (0.2)(0.00) = 0.14$$

The covariance can now be computed using the procedure described in the following table.

Covariance Calculation

Probability	R_A	R_B	Probability $\times [R_A - E(R_A)] \times [R_B - E(R_B)]$
0.3	0.20	0.30	$(0.3)(0.2 - 0.13)(0.3 - 0.14) = 0.00336$
0.5	0.12	0.10	$(0.5)(0.12 - 0.13)(0.1 - 0.14) = 0.00020$
0.2	0.05	0.00	$(0.2)(0.05 - 0.13)(0 - 0.14) = 0.00224$

The covariance of returns for Asset A and Asset B is $0.00336 + 0.00020 + 0.00224 = 0.0058$.

LOS 5.c: Define shortfall risk, calculate the safety-first ratio, and identify an optimal portfolio using Roy's safety-first criterion.

Shortfall risk is the probability that a portfolio value or return will fall below a particular target value or return over a given period.

Roy's safety-first criterion states that the optimal portfolio minimizes the probability that the return of the portfolio falls below some minimum acceptable level. This minimum acceptable level is called the **threshold level**. Symbolically, Roy's safety-first criterion can be stated as follows:

$$\text{minimize } P(R_p < R_L)$$

where:

R_p = portfolio return

R_L = threshold level return

If portfolio returns are normally distributed, then Roy's safety-first criterion can be stated as follows:

$$\text{maximize } \textbf{safety-first ratio}, \text{ which equals } \frac{E(R_p) - R_L}{\sigma_p}$$

The reasoning behind the safety-first criterion is illustrated in **The Safety-First Criterion and Shortfall Risk**. Assume an investor is choosing between two portfolios: Portfolio A, with an expected return of 12% and standard deviation of returns of 18%, and Portfolio B, with an expected return of 10% and standard deviation of returns of 12%. The investor has stated that he wants to minimize the probability of losing money (negative returns). Assuming that returns are normally distributed, the portfolio with the larger safety-first ratio using 0% as the threshold return (R_L) will be the one with the lower probability of negative returns.

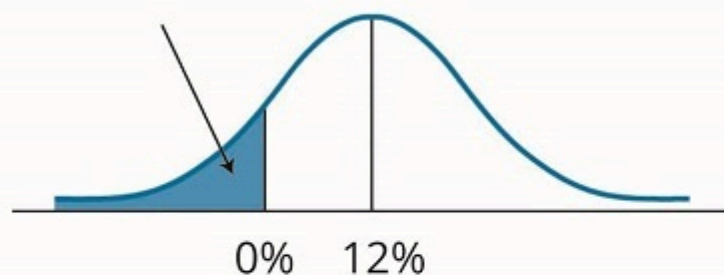
The Safety-First Criterion and Shortfall Risk

A. Normally Distributed Returns

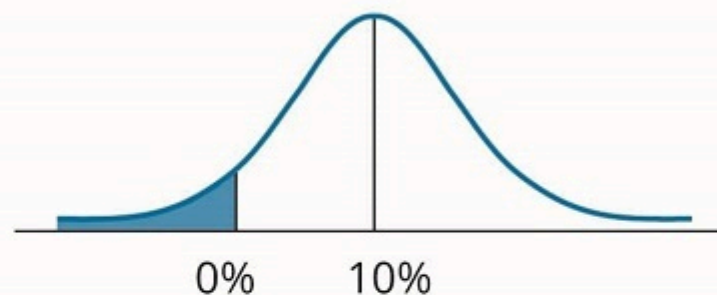
Portfolio A: $E(R) = 12\%$ $\sigma_A = 18\%$

Portfolio B: $E(R) = 10\%$ $\sigma_B = 12\%$

Probability of returns $< 0\%$
– i.e. short fall risk

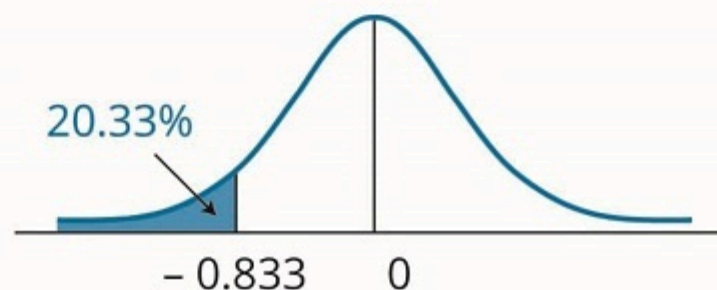
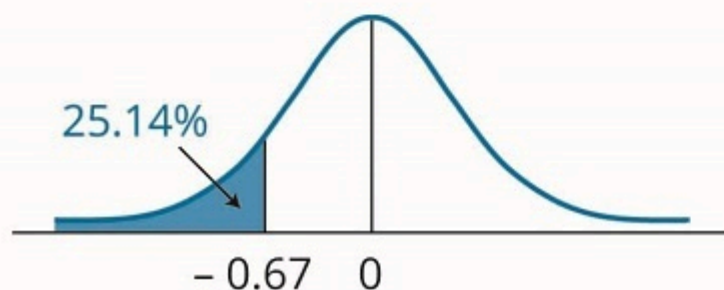


$$SFR_A = \frac{12 - 0}{18} = 0.667$$



$$SFR_B = \frac{10 - 0}{12} = 0.833$$

B. Standard Normal



Panel B of **The Safety-First Criterion and Shortfall Risk** relates the safety-first ratio to the standard normal distribution. Note that the safety-first ratio is the number of standard deviations *below* the mean. Thus, the portfolio with the larger safety-first ratio has the lower probability of returns below the threshold return, which is a return of 0%

in our example. Using a z-table for negative values, we can find the probabilities in the left-hand tails as indicated. These probabilities (25% for Portfolio A and 20% for Portfolio B) are also the shortfall risk for a target return of 0%—that is, the probability of negative returns. Portfolio B has the higher safety-first ratio, which means it has the lower probability of negative returns.

In summary, when choosing among portfolios with normally distributed returns using Roy's safety-first criterion, there are two steps:

Step 1: Calculate the safety-first ratio =
$$\frac{E(R_p) - R_L}{\sigma_p}.$$

Step 2: Choose the portfolio that has the *largest* safety-first ratio.

Example: Roy's safety-first criterion

For the next year, the managers of a \$120 million college endowment plan have set a minimum acceptable end-of-year portfolio value of \$123.6 million. Three portfolios are being considered that have the expected returns and standard deviation shown in the first two rows of the following table. Determine which of these portfolios is the most desirable using Roy's safety-first criterion and the probability that the portfolio value will fall short of the target amount.

Answer:

The threshold return is $R_L = (123.6 - 120) / 120 = 0.030 = 3\%$. The safety-first ratios are shown in the following table. As indicated, the best choice is Portfolio A because it has the largest safety-first ratio.

Roy's Safety-First Ratios

Portfolio	Portfolio A	Portfolio B	Portfolio C
$E(R_p)$	9%	11%	6.6%
σ_p	12%	20%	8.2%

$$\text{SFRatio} \quad 0.5 = \frac{(9-3)}{12} \quad 0.4 = \frac{(11-3)}{20} \quad 0.44 = \frac{(6.6-3)}{8.2}$$

The probability of an ending value for Portfolio A less than \$123.6 million (a return less than 3%) is simply F(-0.5), which we can find on the z-table for negative values. The probability is 0.3085 = 30.85%.

Reading 5: Key Concepts

SCHWESER NOTES - BOOK 1

LOS 5.a

The expected return of a portfolio composed of n assets with weights, w_i , and expected returns, R_i , is:

$$E(R_P) = \sum_{i=1}^n w_i E(R_i) = w_1 E(R_1) + w_2 E(R_2) + \dots + w_n E(R_n)$$

The variance of a portfolio composed of two risky assets, A and B, can be expressed as follows:

$$\sigma_P^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \text{Cov}_{AB}$$

where Cov_{AB} is the expected value of the product of the deviations of the two assets' returns from their respective expected values.

The variance of a two-asset portfolio can also be expressed as follows:

$$\text{Var}(R_P) = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \sigma_A \sigma_B \rho_{A,B}$$

where $\rho_{A,B}$ is the correlation of the two assets' returns.

LOS 5.b

Given the joint probabilities for A_i and B_i , the covariance is calculated as follows:

$$\sum_{i=1}^n P(A_i B_i) [A_i - E(A)] [B_i - E(B)]$$

LOS 5.c

Shortfall risk is the probability that a portfolio's value (or return) will fall below a specific value over a given period.

The safety-first ratio for Portfolio P , based on a target return R_T , is:

$$\text{SFRatio} = \frac{E(R_P) - R_T}{\sigma_P}$$

Greater safety-first ratios are preferred and indicate a smaller shortfall probability. Roy's safety-first criterion states that the optimal portfolio minimizes shortfall risk.